

Anshu Arora

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Education

IIT Bombay, B. Tech in Electrical Engineering Oct 2022 – May 2026

- GPA: **9.35/10.0**, pursuing **Honors** in Electrical Engineering and a **Minor** in Data Science and Artificial Intelligence
- **Key courses**: Large-scale Distributed Optimization and Learning, Advanced Machine Learning, Markov Chains and Queuing Processes, Mathematics of Deep Learning, Online Learning, Optimization models, Digital Signal Processing, Wavelets, Control Systems, Probability and Random Processes, Data Structures and Algorithms, Data Science

Publications

- **Anshu Arora**, Satish Mulleti, "A Lowrate Variable-Bias Integrate-and-Fire Time Encoding Machine", published IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), April 2025
- **Co-author**, "Technical Design Report of Matsya 6D, Autonomous Underwater Vehicle", Naval Engineers Journal, Vol. 136, pp. 161–175, September 2024
- K Yashaswini*, **Anshu Arora***, Satish Mulleti, "A Non-Uniform Quantization Framework for Time-Encoding Machines", under review for IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026
- **Anshu Arora***, K Yashaswini*, Satish Mulleti, "Linear-Bias Time Encoding for Low-Rate Quantized Representation of Bandlimited Signals", currently under review for IEEE Signal Processing Letters **equal contribution*

Scholastic Achievements

- Selected as a recipient of **National University of Singapore Fellowship** for demonstrating research excellence 2025
- Received the **Undergraduate Research Award** for work in signal processing and machine learning 2025
- **Ranked 233** (top 0.02%) in the **JEE Advanced** and **917** (top 0.09%) in the **JEE Mains** (1M+ candidates) 2022
- Awarded the **KVPY Fellowship** conducted by the Department of Science and Technology, **Govt. of India** 2022

Research Projects

Clustering in Federated Reinforcement Learning Mar 2025 – Present

Guides: Prof. Avishek Ghosh and Prof. Pranay Sharma, Computer Science and Engineering, IIT Bombay

- Conducting an in-depth literature review on **federated reinforcement learning** with **TD(0)** and **linear function approximation**, focusing on clustering approaches for **policy evaluation** across distributed agents
- Designed a novel algorithm and rigorously derived **convergence bounds** using a **one-shot clustering approach** followed by collaboration of agents in a cluster through aggregation of local TD(0) update directions
- Analyzed the trade-off between misclassifications of agents and **linear speed-up** in convergence due to collaboration

Robustness in Federated Learning Sep 2025 – Present

Guide: Prof. Pranay Sharma, Center for Machine Intelligence and Data Science, IIT Bombay

- Investigating the existing research on Byzantine agents and robust aggregation methods in multi-agent systems
- Modeling the notion of **stealth in adversarial attacks**, as a multi-objective optimization problem
- Evaluating the attack success rate using the **MGDA algorithm**, to converge to **Pareto stationary** points

Consensus on Lie Groups with Unreliable Networks Jun 2025 – Present

Guide: Prof. Ravi Banavar, Systems and Control Engineering, IIT Bombay

- Examining consensus dynamics on **Lie Groups** under lossy connectivity in **multi-agent information networks**
- Analyzing the relation between the **Vicsek model** and **first-order kinematic model** of consensus, employing a **switching function analysis** and local strong convexity assumptions on Lie groups with a **bi-invariant metric**
- Characterizing the **domain of initial conditions** that permit consensus despite communication disruptions

Time Encoding Machines (TEMs) May 2024 – Nov 2025

Guide: Prof. Satish Mulleti, Electrical Engineering, IIT Bombay

- Designed a time-varying bias to closely track the input signal, leading to a significant **reduction in oversampling**
- Derived **mathematical transformations** between the probability density function of the amplitude and time encodings, to design an efficient **non-uniform quantizer**, significantly reducing the information throughput
- **Bypassed the Bernstein inequality** in the operator-based reconstruction mechanism of TEMs, to remove the bottleneck of sampling above the Nyquist rate, without compromising on the perfect reconstruction guarantee

Key Projects

Classification using Spiking Neural Networks (SNNs)	Oct 2025 – Dec 2025
- Used SNNs with spike time dependent plasticity as the learning mechanism to perform unsupervised MSNIST dataset classification, based on the Leaky Integrate-and-Fire neuron model architecture - Developed a supervised MNIST classifier using snnTorch with backpropagation through time training	
Inference in Machine Learning	Jan 2025 – May 2025
- Evaluated message passing algorithms for computing marginals and MAP assignments on graphical models - Implemented denoising diffusion probabilistic models for unconditional and conditional generation, analyzing the effects of diffusion steps, noise schedules, and classifier-free guidance on sample quality - Deployed MCMC sampling algorithms for energy-based models and implemented Gaussian Process regression	
EMG Controlled Prosthetic Arm	Jan 2025 – May 2025
- Placed top 3 among 40+ teams for developing a myoelectric system to classify 3 motions from EMG signals - Applied signal processing for feature extraction and implemented an SVM-based model on Neuropype software	
Optimization of Traffic Light Timings	Nov 2024 – Dec 2024
- Built linear programming models on a time-space network with Pyomo to minimize the commuter wait times - Analysed reinforcement learning algorithms to optimize timings under stochastic intersection conditions	
Wavelets in Magnetic Resonance Imaging	Jul 2024 – Nov 2024
- Designed a CNN-based deep learning architecture using curvelets as features for tri-segmentation of the brain - Incorporated high-resolution time-frequency features into the model, achieving an IoU score of 0.71	

Positions of Responsibility

Mentorship Program Head , Electrical Dept. – Student Mentorship Program	March 2025 – Present
- One of 3 departmental student representatives leading 1K+ students, entrusted with advancing academic excellence - Leading a 3-tier mentorship network of 5 subgroups and 50+ mentors to support 200+ undergraduates	
Captain , Women’s Basketball Team – IIT Bombay Sports	Aug 2024 – Present
- Led the team to its first ever Gold Medal at the 57th Inter-IIT tournament, from 20+ participating IITs - Coordinating fitness initiatives, structured team practice schedules, fostering strong team cohesion and bonding	
Chief Executive – Autonomous Underwater Vehicle Team	Aug 2025 – Present
- Part of a 50+ member student robotics team developing Matsya, an autonomous underwater vehicle integrating visual, inertial, acoustic, and depth sensing with thruster and servo actuation for realistic naval missions - Leading the business division, responsible for procuring sponsorships and performing outreach activities - Achieved an overall 4th rank in the international competition RoboSub ’24 at California among 40+ teams	

Experiences

Volunteering Activities	
- Volunteered with Ummeed NGO, teaching mathematics to disadvantaged students over the summers - Worked with young girls to take up STEM as a part of Women in Science and Engineering program - Organized inclusive events for people with disabilities at GODS Foundation and Old Saints Home	
Mentoring Experience	
- Served as a teaching assistant for 2 semesters, for Linear Algebra and Classical & Quantum Physics - Selected in a team of 35+ mentors from 175+ applicants to provide academic guidance to 12 sophomores - One of 19 pre-final year mentors, selected from 1400+ peers to provide mentorship to 12 freshmen	
Extracurricular Activities	
- Appointed Junior Consultant at IITB’s Core Consulting Group in recognition of strong communication skills - Secured 2nd place in the Annual Freshers Debate Open, competing against 1400+ participants - Participated in the State level Basketball tournament of Tamil Nadu and represented the Chennai District - Awarded with distinction grade in Level 5 Keyboard Certification, by the Trinity College London	